Project Proposal: Eco-TMS

Edge-AI & Fuzzy Inference Driven Traffic Decision-Support

Submission for NESFIC-D-22: AI-Based Smart Transport Monitoring

1 Executive Summary & Problem Alignment

Urban traffic monitoring currently relies on fragmented, manual observations or rigid, centralized systems that fail to account for non-lane-based, mixed-vehicle dynamics (e.g., high volumes of two-wheelers and auto-rickshaws) and localized environmental disruptions. Eco-TMS addresses the NESFIC-D-22 mandate by deploying a decentralized, low-cost edge computing network that provides continuous spatial coverage, combining lightweight computer vision with real-time weather sensing to dynamically manage urban bottlenecks.

2 Hardware & Edge Integration (The Sensor Node)

Instead of relying on expensive, high-bandwidth PTZ cameras streaming to a central server, Eco-TMS utilizes localized hardware to process data at the intersection level.

- **Edge Processing Units:** Low-cost microprocessors analyze camera feeds locally, extracting only numerical metadata (vehicle counts, trajectory, speed) to eliminate heavy video transmission and reduce latency.
- **Environmental Sensor Array:** Intersections are equipped with ESP32 microcontrollers wired to DHT (humidity/temperature) and optical sensors to detect visibility drops (fog) or road surface waterlogging.
- **Low-Power Communication:** The ESP32 array transmits this lightweight, structured metadata to the central traffic decision system using MQTT protocols, ensuring continuous coverage even during low-bandwidth scenarios.

3 Machine Learning & Backend Architecture

The software stack bridges real-time localized adjustments with long-term predictive analytics.

- **Mixed-Traffic Computer Vision:** A YOLO-based object detection model (implemented via OpenCV) is trained specifically on Indian urban traffic patterns. Rather than just counting discrete cars, the model calculates spatial density, effectively tracking motorcycles weaving through gaps and providing a true congestion metric.
- **Predictive Cloud Backend:** A Python backend utilizing libraries like `scikit-learn` aggregates the metadata across all city nodes. It identifies recurring bottleneck patterns and predicts peak congestion zones before they form. This feeds into a dashboard for city planners, allowing for proactive traffic diversion or dynamic public transit rerouting.

4 The Fuzzy Inference Engine (Signal Logic)

Rigid thresholds fail in mixed traffic. A fuzzy logic controller takes continuous inputs and handles the ambiguity of real-world congestion seamlessly.

Antecedents (Inputs)

1. **Vehicle Density (from Camera):** *Low, Medium, High* (Measured as the percentage of road space occupied).
2. **Average Approach Speed (from Camera):** *Slow, Moderate, Fast.*
3. **Weather Severity (from ESP32):** *Clear, Mild, Severe* (Derived from the rain and visibility sensors).

Consequents (Outputs)

1. **Green Light Extension:** *Short, Standard, Long.*
2. **Yellow Light Duration:** *Standard, Extended.*

Sample Fuzzy Rule Base

- **Rule 1 (The Baseline):** IF Density is *High* AND Weather is *Clear*, THEN Green Light is *Long* AND Yellow Light is *Standard.*
- **Rule 2 (The Weather Adjustment):** IF Density is *High* AND Weather is *Severe*, THEN Green Light is *Long* AND Yellow Light is *Extended.* (Prevents hard braking on wet asphalt).
- **Rule 3 (The Flow Optimization):** IF Density is *Low* AND Speed is *Fast* AND Weather is *Clear*, THEN Green Light is *Short.*

5 Implementation Plan

- **Phase 1 (Simulation):** Build a Python simulation using OpenCV on pre-recorded traffic footage from a busy local intersection, running alongside a simulated `skfuzzy` control loop to demonstrate wait-time reduction.
- **Phase 2 (Hardware PoC):** Assemble a single ESP32 sensor node to demonstrate real-time environmental telemetry being published to a cloud dashboard via MQTT.

6 Value Proposition for NESFIC

Scalability: By keeping the processing at the edge and utilizing low-cost microcontrollers, the system can be deployed across a massive geographical area at a fraction of the cost of traditional smart-city infrastructure.
Resilience: The hybrid approach ensures that if a camera is obscured, the environmental sensors and historical predictive models continue to optimize traffic flow safely.